6. A girl standing 6·0 m from a convex mirror sees her virtual image 1·0 m behind the mirror. The focal length of the mirror is : *1*

(A) 6·0 m (B) 1·2 m

(C) 2·4 m (D) 0·8 m

CBSE 26C

6. A girl standing 6·0 m from a convex mirror sees her virtual image 1·0 m behind the mirror. The focal length of the mirror is : *1*

(A) 6·0 m (B) 1·2 m

(C) 2·4 m (D) 0·8 m

CBSE 26C

(B) 1.2 m

20. What is meant by linear magnification of an image formed by a spherical mirror ? Write an expression for the magnification produced by a concave mirror in terms of the object distance and the image distance. CBSE 26C 2

20. What is meant by linear magnification of an image formed by a spherical mirror ? Write an expression for the magnification produced by a concave mirror in terms of the object distance and the image distance. CBSE 26C 2

Linear magnification is the ratio of the size of image to the size of object. Alternatively: $$m = \frac{h_I}{h_O} = \frac{\text{Size of image}}{\text{Size of object}}$$	1
$$m = -\frac{v}{u} = -\frac{\text{Image distance}}{\text{Object distance}}$$	1

24. (a) The distance of an object from a concave mirror is increased from 12 cm to 15 cm. It is observed that the new image becomes $\left(\frac{2}{5}\right)^{\text{th}}$ the size of the initial image. Calculate the focal length of the mirror.

(b) A light ray is incident on the horizontal upper face of a glass cube making an angle of 45° with the face. While emerging, the ray just grazes the vertical face of the cube. Find the refractive index of the glass. CBSE 26C 3

(a)

For $u = -12\ cm$

$$m_1 = \frac{h_1}{h_o} = \frac{f}{f+12} \quad \ldots(1)$$ ½

For $u = -15\ cm$

$$m_2 = \frac{h_2}{h_o} = \frac{f}{f+15} \quad \ldots(2)$$ ½

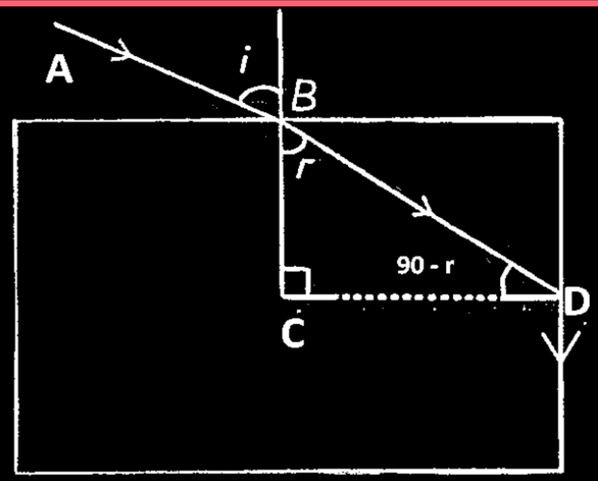
Dividing (1) by (2)

$$\frac{m_1}{m_2} = \frac{h_1}{h_2} = \frac{f+15}{f+12}$$

$$h_2 = h_1\left(\frac{2}{3}\right)$$

On solving,

$$f = -10\ cm$$ ½



At air-glass interface,

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin 45^\circ}{\sin r}$$

$$\sin r = \frac{\sin 45^\circ}{n} = \frac{1}{\sqrt{2}\,n} \quad \ldots(1)$$ ½

At glass-air interface, light suffers Total Internal Reflection (TIR) ($i_C = 90°\text{-}r$).

$$n = \frac{1}{\sin i_C}$$

$$n = \frac{1}{\sin(90^\circ - r)} = \frac{1}{\cos r}$$

$$n = \frac{1}{cosr}$$

$$\cos r = \frac{1}{n} \quad \ldots(2)$$ ½

Squaring and adding (1) and (2)

$$1 = \frac{1}{2n^2} + \frac{1}{n^2}$$

On solving,

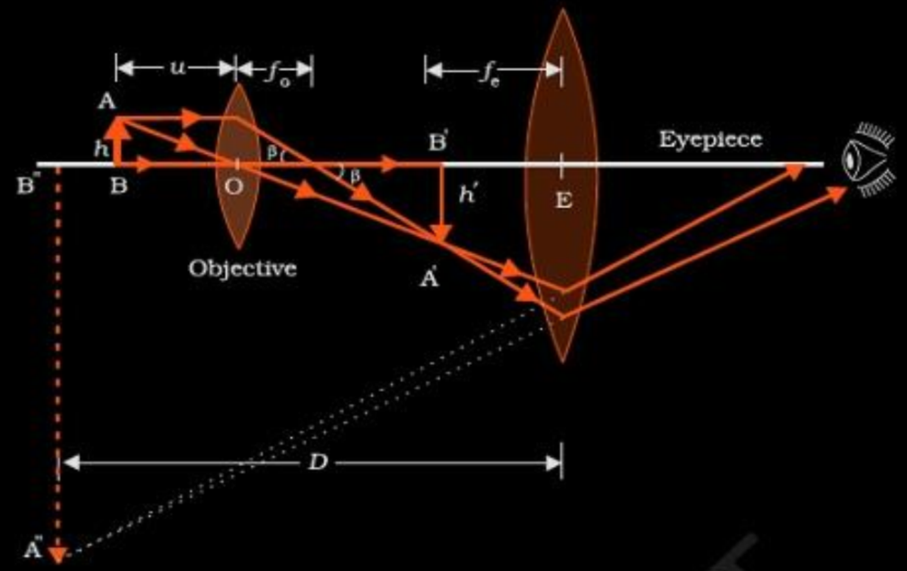
$$n = \sqrt{\frac{3}{2}}$$ ½

(b) (i) With the help of a ray diagram, describe the construction and working of a compound microscope.

(ii) (I) The real image of an object placed between f and 2f from a convex lens can be seen on a screen placed at the image location. If the screen is removed, is the image still there ? Explain.

(II) Plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? Explain. CBSE 2026 5

(i)



1

Construction: It consists of two convex lenses, objective lens with small aperture placed near the object and the eye piece with large aperture placed near the eye. Both lenses have small focal length. These lenses are placed coaxially in a narrow tube with rack and pinion arrangement.

1

Working:
The lens near to the object called objective forms a real, inverted & magnified image of the object. This serves as the object for eye piece which produces the final image, enlarged & virtual.

1

(ii) I Yes ½

Real image is formed by actual intersection of rays. ½

II Yes ½

If the object is virtual. ½

Note: 1. If students do not write construction award full marks for well-labelled diagram representing objective lens, eye piece, along with image position at near point or normal adjustment.

2. Deduct ½ mark if arrows are not drawn.

19. A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object. *2*

CBSE 2026

19. A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object. *2*

CBSE 2026

Given: $R = -16$ cm (concave mirror)

$m = -\frac{v}{u} = -2$

$v = 2u$ ½

Now for concave mirror,

$f = -8\text{cm}, v = 2u, u = ?$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ ½

$\frac{1}{2u} + \frac{1}{u} = \frac{1}{-8}$ ½

$\frac{1+2}{2u} = \frac{1}{-8}$

$\frac{2u}{3} = -8$

$u = -\frac{24}{2} = -12\text{cm}$

$\therefore u = -12\text{cm}$ ½

18. (a) A ray of light travelling along the principal axis is incident on a spherical mirror and reflected back along the same path. Explain, why ? **2**

OR

(b) Write all the sign convention followed for a convex lens. **2**

CBSE 2026

18. (a) A ray of light travelling along the principal axis is incident on a spherical mirror and reflected back along the same path. Explain, why ? **2**

OR

(b) Write all the sign convention followed for a convex lens. **2**

CBSE 2026

When a ray of light travelling along the principal axis is incident on a spherical mirror, then incident light is normal to the surface of spherical mirror, so it reflects back along the same path. **2**

Alternatively,

The angle of incidence along the principal axis is zero therefore from the laws of reflection, the angle of reflection will be zero. Thus, it reflects back along the same path.

½ ×4

1. All the distances are measured from the optical center of the lens.
2. Distances measured along the direction of the incident ray are taken as positive.
3. Distance measured opposite to the direction of incident ray are taken as negative.
4. Height above the principal axis is taken as positive and height below the principal axis is taken as negative.

21. A 5 cm long pencil is placed along the principal axis of a concave mirror of focal length 20 cm such that its nearest end is at a distance of 25 cm from the mirror. Calculate the length of the image of the pencil. CBSE 2026 **2**

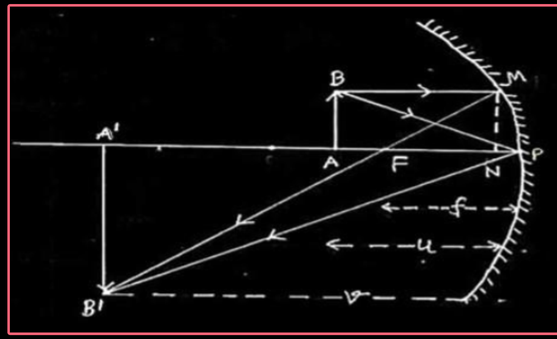
21. A 5 cm long pencil is placed along the principal axis of a concave mirror of focal length 20 cm such that its nearest end is at a distance of 25 cm from the mirror. Calculate the length of the image of the pencil. CBSE 2026 **2**

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	$^1/_2$
$\frac{1}{v_1} = \frac{1}{f} - \frac{1}{u_1}$	
$= \frac{1}{(-20)} - \frac{1}{(-25)}$ (For near end of the pencil)	
$v_1 = -100\text{cm}$	$^1/_2$
$\frac{1}{v_2} = \frac{1}{(-20)} - \frac{1}{(-30)}$ (For far end of the pencil)	
$v_2 = -60$ cm	$^1/_2$
Length of the image L = $\lvert v_1 \rvert - \lvert v_2 \rvert = 40\text{cm}$	$^1/_2$

(a) Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the centre of curvature. Using this diagram, derive the mirror formula.

(b) A concave mirror produces a two times magnified virtual image of an object kept 10 cm in front of it. Calculate the focal length of the mirror.

CBSE 2026



1

ΔBAP and $\Delta B'A'P$ are similar then

$$\frac{BA}{B'A'} = \frac{AP}{A'P} \qquad \text{----------(1)}$$ ½

$\because BA = MN$

ΔMNF and $\Delta B'A'F$ are similar then

$$\frac{MN}{B'A'} = \frac{NF}{A'F}$$

$$\frac{BA}{B'A'} = \frac{NF}{A'F} = \frac{PF}{A'F}$$ (N is very close to P and BA = MN) --------(2) ½

From equation (1) and (2)

$$\frac{AP}{A'P} = \frac{PF}{A'F}$$

$$\frac{-u}{-v} = \frac{-f}{-v+f}$$ ½

$$-uv + uf = -vf$$

$$\frac{-uv}{uvf} + \frac{uf}{uvf} = \frac{-vf}{uvf}$$

$$\frac{-1}{f} + \frac{1}{v} = \frac{1}{-u}$$

$$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$ ½

Note: Award full marks if student derives mirror formula by using any other justified method

(b) $m = \frac{-v}{u}$ ½

$m = 2$

$\because v = -2u \Rightarrow v = -2(-10) = 20\ cm$ ½

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{20} - \frac{1}{10} = \frac{1}{f}$ ½

$\frac{1-2}{20} = \frac{1}{f}$

$f = -20\ cm$ ½

20. (a) Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus."

CBSE 2025

2

20. (a) Using the mirror equation and the formula of magnification, deduce that "the virtual image produced by a convex mirror is always diminished in size and is located between the pole and the focus." CBSE 2025 *2*

Let, $u = nf$

From the Mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{½}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{f} + \frac{1}{nf}$$

$$v = \frac{nf}{n+1}$$

$n = +ve \quad \therefore v < f$ ½

$$m = \frac{-v}{u} = \frac{1}{n+1}$$ ½

m is always positive & less than 1. ½

Note: **Award full credit if correctly concluded by any other method.**

25. (a) A concave mirror has radius of curvature 20 cm. Calculate the distance of an object from the mirror so as to form an image of magnification – 2. Also find the location of the image.

(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object ? Justify your answer.

CBSE 2025

3

25. (a) A concave mirror has radius of curvature 20 cm. Calculate the distance of an object from the mirror so as to form an image of magnification – 2. Also find the location of the image.

(b) If the silver coating around the centre of a concave mirror is removed, will the mirror still form the image of an object ? Justify your answer.

CBSE 2025 *3*

(a) $m = -\frac{v}{u}$	½
$-2 = -\frac{v}{u}$	
$v = 2u$	
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	
$\frac{1}{-10} = \frac{1}{2u} + \frac{1}{u}$	½
$u = -15\,\text{cm}$	½
$v = -30\,\text{cm}$	½
(b) Yes, same image is formed with reduced intensity, because reflecting area get reduced and laws of reflection still hold good for remaining part of the mirror.	1

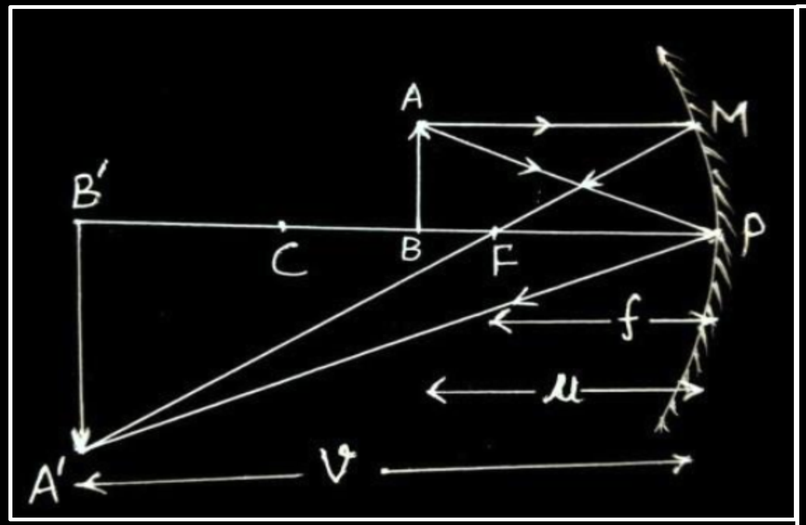
(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

(ii) Use mirror formula to deduce that a convex mirror always produces a virtual image of an object kept in front of it.

(ii)	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$	½
	$u<0 \ , \ f>0$	
	$\frac{1}{v}+\frac{1}{(-u)}=\frac{1}{f}$	½
	$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}$	½
	$\frac{1}{v}$ is positive	
	$\therefore$ v is positive $\Rightarrow$ virtual image	½

26. Draw a ray diagram showing the image formation when a concave mirror produces a real, inverted and magnified image of an object and hence obtain the mirror formula. CBSE 2025 *3*

26. Draw a ray diagram showing the image formation when a concave mirror produces a real, inverted and magnified image of an object and hence obtain the mirror formula. CBSE 2025 *3*



Note: Please deduct ½ mark of this diagram if not showing arrows with the rays.

In similar triangles

$\Delta A'B'F$ *and* ΔMPF — 1

$$\frac{A'B'}{MP}=\frac{B'F}{FP}$$

or $$\frac{A'B'}{AB}=\frac{B'F}{FP} \quad (\because MP=AB) \quad \text{---------(1)}$$ — ½

In similar triangles $\Delta A'B'P$ *and* ΔABP

$$\frac{A'B'}{AB}=\frac{PB'}{PB} \quad \text{-------(2)}$$ — ½

from equation (1) and (2)

$$\frac{B'F}{FP}=\frac{PB'}{PB}$$

$$\frac{PB'-PF}{FP}=\frac{PB'}{PB}$$

$$\frac{(-v)-(-f)}{(-f)}=\frac{(-v)}{(-u)}$$ — ½

on solving we get

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$ — ½

18. (a) The real image of an object is formed at a distance twice the distance of the object from a concave mirror of focal length 10 cm. Find the distance between the object and the image. **2**

CBSE 2025

18. (a) The real image of an object is formed at a distance twice the distance of the object from a concave mirror of focal length 10 cm. Find the distance between the object and the image. **2**

CBSE 2025

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad ½$$

$u=-x,\ \ v=-2x$ ½

$$\frac{1}{-10}=\frac{1}{-x}+\frac{1}{-2x}=-\frac{3}{2x} \qquad ½$$

$x=15\,\text{cm}$

Distance between object and image,
d = 2x- x = 30- 15 = 15cm ½

9. The magnification produced by a spherical mirror is – 2·0. The mirror used and the nature of the image formed will be

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

CBSE 2025

9. The magnification produced by a spherical mirror is – 2·0. The mirror used and the nature of the image formed will be

(A) Convex and virtual

(B) Concave and real

(C) Concave and virtual

(D) Convex and real

CBSE 2025

(B) Concave and real

32. (a) (i) A thin pencil of length (f/4) is placed coinciding with the principal axis of a mirror of focal length f. The image of the pencil is real and enlarged, just touches the pencil. Calculate the magnification produced by the mirror.

CBSE 2025

(i) As the pencil lies between f and 2f such that one end of the pencil coincides with 2f.

Position of the other end (u) $= -\left(2f - \frac{f}{4}\right) = -\frac{7f}{4}$ ½

Magnification (m) $= \frac{f}{f - u}$ ½

$= \frac{-f}{-f - \left(-\frac{7f}{4}\right)}$ ½

$m = -\frac{4}{3}$ 1

Alternatively: -

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{4}{7f} = -\frac{1}{f}$

$\frac{1}{v} = -\frac{1}{f} + \frac{4}{7f}$

$v = -\frac{7f}{3}$

$m = -\frac{v}{u} = -\frac{4}{3}$

9. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $\frac{5}{3}$, its focal length will become **1**

(A) $\frac{5}{3}$ f

(B) $\frac{3}{5}$ f

(C) $\frac{2}{3}$ f

(D) f

CBSE 2025

9. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $\frac{5}{3}$, its focal length will become **1**

(A) $\frac{5}{3}$ f

(B) $\frac{3}{5}$ f

(C) $\frac{2}{3}$ f

(D) f

CBSE 2025

(D) f

(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image.

CBSE 2024

(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image.

CBSE 2024

(ii)

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$

$$v=\frac{uf}{u-f}$$

Following new cartesian sign conversion

$$v=\frac{(-u)(-f)}{-u-(-f)}$$

$$v=\frac{uf}{f-u} \qquad \text{as } f>u$$

1

v is +ve, So image is virtual.

$$m=-\frac{v}{u}=\frac{f}{f-u}>1$$ i.e. Enlarged image

1

19. The radius of curvature of a convex mirror is 30 cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.

CBSE 2024

19. The radius of curvature of a convex mirror is 30 cm. It forms an image of an object which is half the size of the object. Find the separation between the object and the image.

CBSE 2024

$$m = -\frac{v}{u} = \frac{h_I}{h_O} = \frac{1}{2}$$ ½

$$u = -2v$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$ ½

$$\frac{1}{15} = \frac{1}{v} - \frac{1}{2v}$$

On solving

$$|v| = 7.5\,\text{cm}$$

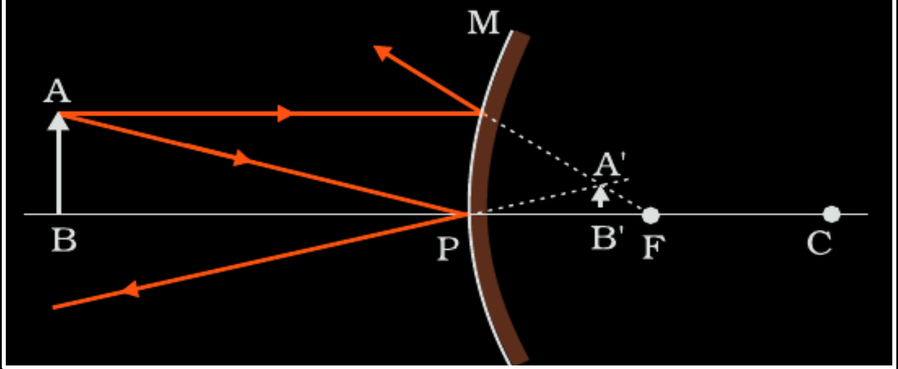
$$|u| = +15.0\,\text{cm}$$

Separation = 15.0 + 7.5 ½

= 22.5 cm ½

32. (a) (i) Draw a ray diagram for the formation of the image of an object by a convex mirror. Hence, obtain the mirror equation.

CBSE 2024



For paraxial rays MP can be considered to be a straight line perpendicular to CP,

Therefore right angled triangles $A'B'F$ and MPF are similar

$$\frac{B'A'}{PM} = \frac{B'F}{FP}$$

1

Or $\frac{B'A'}{BA} = \frac{B'F}{FP}$ ($\because$ PM = AB) ----------(1) ½

Since $\angle$ APB = $\angle$ $A'PB'$, the right angled triangles $A'PB'$ and ABP are also similar

Therefore , $\frac{B'A'}{BA} = \frac{B'P}{BP}$ ---------- (2) ½

Comparing eq (1) and (2), we get

$$\frac{B'F}{FP} = \frac{B'P}{BP}$$

$$\frac{PF - PB'}{FP} = \frac{B'P}{BP}$$

Using sign convention

PF = f, PB' = +v , PB = -u ½

on solving $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ ½

19. Explain the following : *2*

(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? CBSE 2024

19. Explain the following : *2*

(b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? CBSE 2024

(b) Yes, Rays converging to a point behind a plane or convex mirror are reflected to a point in front of the mirror on a screen	½ ½

14. *Assertion (A) :* Plane and convex mirrors cannot produce real images under any circumstance.

Reason (R) : A virtual image cannot serve as an object to produce a real image.

CBSE 2024

14. *Assertion (A) :* Plane and convex mirrors cannot produce real images under any circumstance.

Reason (R) : A virtual image cannot serve as an object to produce a real image.

CBSE 2024

(D) Both Assertion (A) and Reason (R) are false.

18. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm. Find the (i) position of the image formed and (ii) magnification of the image. CBSE 2024 *2*

18. An object is placed 30 cm in front of a concave mirror of radius of curvature 40 cm. Find the (i) position of the image formed and (ii) magnification of the image. CBSE 2024 *2*

(i) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} + \frac{1}{-30} = \frac{1}{-20}$	½
On solving v = - 60 cm	½
(ii) m = $-\frac{v}{u}$	½
$= -(\frac{-60}{-30}) = -2$	½

(ii) An object is kept 20 cm from a convex mirror of radius of curvature 60 cm. Find the position of the image formed. Will the image be real or virtual ?

CBSE 2024

(ii) An object is kept 20 cm from a convex mirror of radius of curvature 60 cm. Find the position of the image formed. Will the image be real or virtual ?

CBSE 2024

(ii) u= -20cm

$$f = \frac{R}{2} = 30\text{cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \left(-\frac{1}{20}\right) = \frac{1}{30}$$ ½

$$\frac{1}{v} = \frac{1}{30} + \frac{1}{20} = \frac{3+2}{6} = \frac{5}{60}$$ ½

$$v = 12\,\text{cm}$$ ½

Virtual Image. ½

33. (a) (i) Draw a ray diagram to show the formation of image by a concave mirror when the object is placed between f and 2f. Using this diagram, derive the mirror equation.

(ii) An object is kept 30 cm in front of a convex mirror of focal length 12 cm. Find the nature and position of the image formed. *5*

CBSE 2023 C

From $\triangle A'B'C' \sim \triangle DPF$

$$\frac{B'A'}{BA} = \frac{B'F}{FP} (\because PD = AB) \quad \text{---------(1)}$$

Form $\triangle A'B'P \sim \triangle ABP$ ½

$$\frac{B'A'}{BA} = \frac{B'P}{BP} \quad \text{---------(2)}$$

½

Comparing equation (1) and (2)

$$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$$

Here B'P = - v

FP = -f

BP = - u

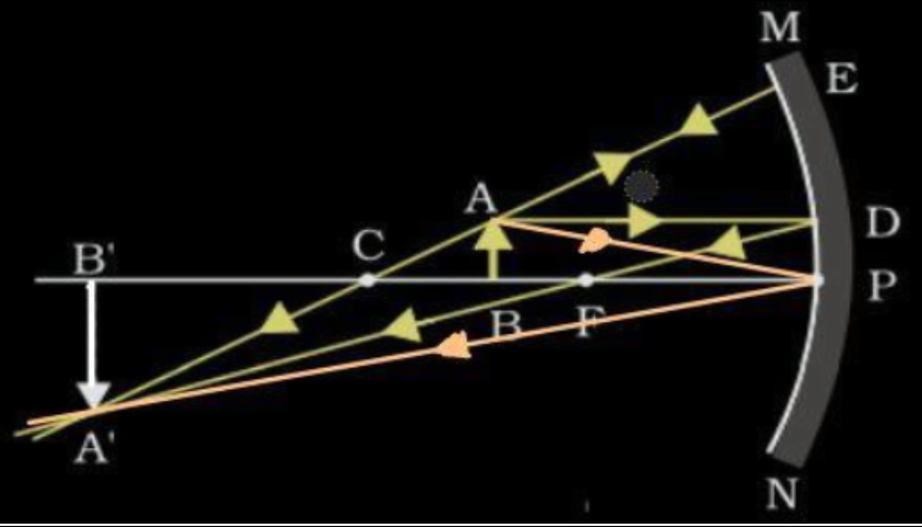
$$\frac{-v + f}{-f} = \frac{-v}{-u}$$ ½

$$\frac{v - f}{f} = \frac{v}{u}$$

$$\frac{v}{f} = 1 + \frac{v}{u}$$

On solving we get

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$ ½



(ii) u= -30 cm

f= + 12 cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{12} + \frac{1}{30} = \frac{7}{60}$$ 1

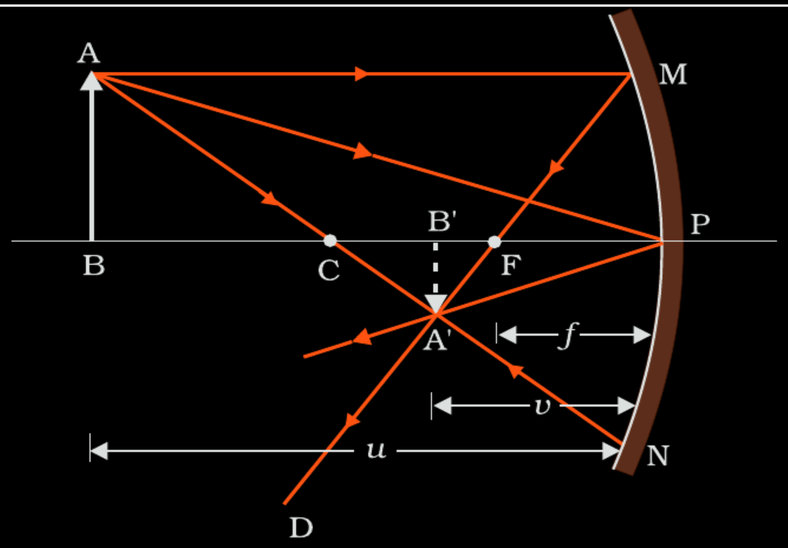
v= $\frac{60}{7}$ cm ½

Since v is positive image is virtual and erect.

31. (a) Draw a ray diagram for formation of a real and diminished image of an object kept in front of a concave mirror. Hence derive the mirror equation.

(b) A concave mirror of focal length 10 cm produces a real image which is 3 times the size of the object. Find the distance of the object from the mirror. *5*

CBSE 2023 C



1

$\Delta A'B'F \sim \Delta MPF$

$\frac{B'A'}{BA} = \frac{B'F}{FP}$ $(\because PM = AB)$ _____(1) ½

Also, $\Delta A'B'P \sim \Delta ABP$

$\frac{B'A'}{BA} = \frac{B'P}{BP}$ _____(2) ½

Comparing eq. (1) and (2)

$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{FP}$ ½

$B'P = -v$

$FP = -f$

$BP = -u$

On solving we get , $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ ½

(b)

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ ½

$\frac{1}{u} = \frac{1}{f} - \frac{1}{v}$ ½

$-\frac{v}{u} = -3$

$v = 3u$ ½

$\frac{1}{u} = \frac{1}{f} - \frac{1}{3u}$

$\frac{1}{u} + \frac{1}{3u} = \frac{1}{f}$

$\frac{4}{3u} = \frac{1}{f}$ ½

$u = \frac{4f}{3} = -\frac{40}{3} cm$ ½

10. An object is placed at a distance of 10 cm in front of a concave mirror of radius of curvature 15 cm. The nature and position the image formed is :

(a) real and inverted, 30 cm in front of the mirror

(b) real and inverted, 15 cm in front of the mirror

(c) virtual and erect, 30 cm behind the mirror

(d) virtual and erect, 15 cm behind the mirror

CBSE 2023

10. An object is placed at a distance of 10 cm in front of a concave mirror of radius of curvature 15 cm. The nature and position the image formed is :

(a) real and inverted, 30 cm in front of the mirror

(b) real and inverted, 15 cm in front of the mirror

(c) virtual and erect, 30 cm behind the mirror

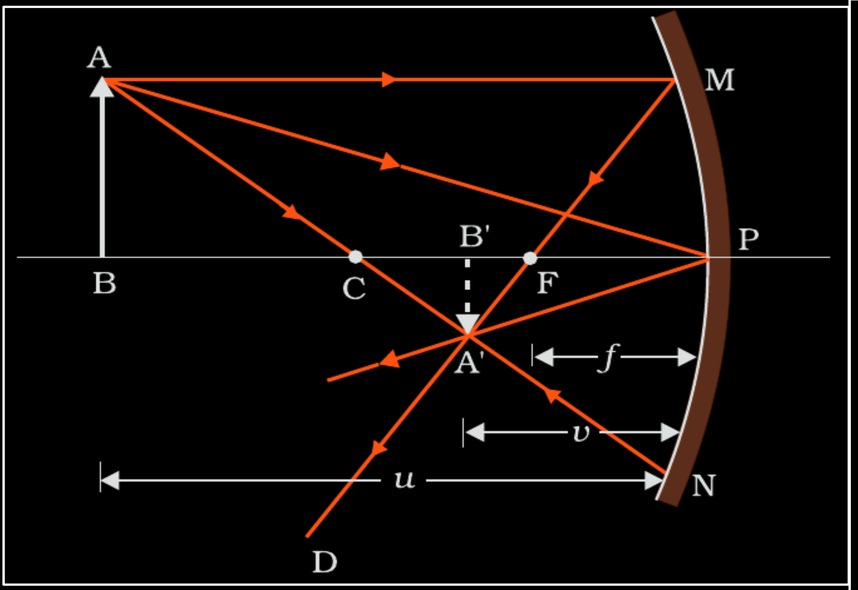
(d) virtual and erect, 15 cm behind the mirror

CBSE 2023

(a) real and inverted, 30 cm in front of the mirror

31. (a) (i) Draw a ray diagram showing the formation of a real image of an object placed at a distance 'u' in front of a concave mirror of radius of curvature 'R'. Hence, obtain the relation for the image distance 'v' in terms of u and R.

CBSE 2023



From Fig. the two right-angled triangles A′B′F and MPF are similar. (For paraxial rays, MP can be considered to be a straight line perpendicular to CP.) Therefore,

$$\frac{B'A'}{PM} = \frac{B'F}{FP}$$

$$\text{or } \frac{B'A'}{BA} = \frac{B'F}{FP} \quad (\because PM = AB)$$

-----(i) ½

Since ∠ APB = ∠ A′PB′, the right angled triangles A′B′P and ABP are also similar. Therefore,

$$\frac{B'A'}{BA} = \frac{B'P}{BP}$$

----(ii) ½

Comparing equations (i) and (ii)

$$\frac{B'F}{FP} = \frac{B'P - FP}{FP} = \frac{B'P}{BP}$$

-----(iii) ½

B′ P = – v, FP = – f, BP = – u;

Using these in Eq.(iii) we get $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$ ½

Alternatively:- If the result derived by any other method, full credit to be given.

9. For a concave mirror of focal length 'f', the minimum distance between the object and its real image is :

(a) zero

(b) f

(c) 2f

(d) 4f

CBSE 2023

9. For a concave mirror of focal length 'f', the minimum distance between the object and its real image is :

(a) zero

(b) f

(c) 2f

(d) 4f

CBSE 2023

(a) Zero

(ii) A concave mirror of focal length 12 cm forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.

CBSE 2023

(ii) A concave mirror of focal length 12 cm forms a three times magnified virtual image of an object. Find the distance of the object from the mirror.

CBSE 2023

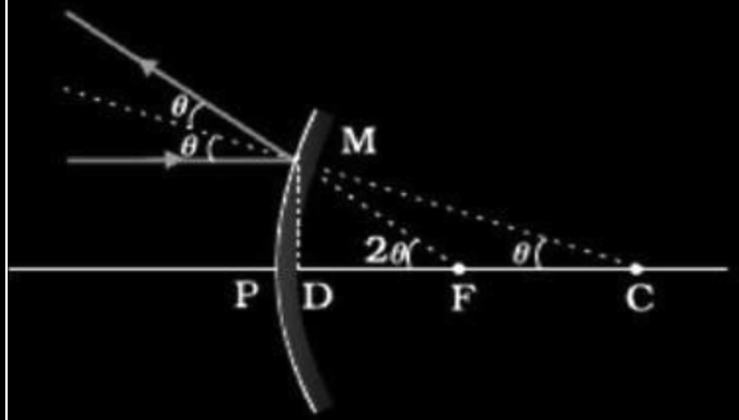
(ii) $m = +3$, f = -12 cm , $u = ?$	
$m = -\frac{v}{u} = 3 \Rightarrow v = -3u$	½
using mirror formula	
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	½
$\frac{1}{-3u} + \frac{1}{u} = \frac{1}{-12}$	½
u = - 8 cm	½

35. (a) Define the term 'focal length of a mirror'. With the help of a ray diagram, obtain the relation between its focal length and radius of curvature.

CBSE 2020

(a) **Focal length of mirror**: It is the distance of the point from the pole of mirror through which ray of light moving parallel to its principle axis passes (or appear to come from). 1

Alternatively: It is half of the distance of its centre of curvature from the pole of a mirror.



Let C be the centre of curvature of mirror, MD be the perpendicular from M to the principal axis.

$\angle MCP = \theta$ *and* $\angle MFP = 2\theta$

$$\tan\theta = \frac{MD}{CD} \quad , \quad \tan 2\theta = \frac{MD}{FD} \qquad (1)$$ ½

For small angles, $\tan\theta \approx \theta$ and $\tan 2\theta \approx 2\theta$ ½

From equation 1, $\dfrac{MD}{FD} = 2\dfrac{MD}{CD}$

$$FD = \frac{CD}{2} \qquad ----equation\ (2)$$

For small θ, the point D is very close to the point P

$\therefore$ FD $\simeq$ FP $= f$ and CD $\simeq$ CP $= R$

$\therefore$ from equation 2, we get f $= \dfrac{R}{2}$ ½

34. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror. CBSE 2020 **3**

34. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror. CBSE 2020 **3**

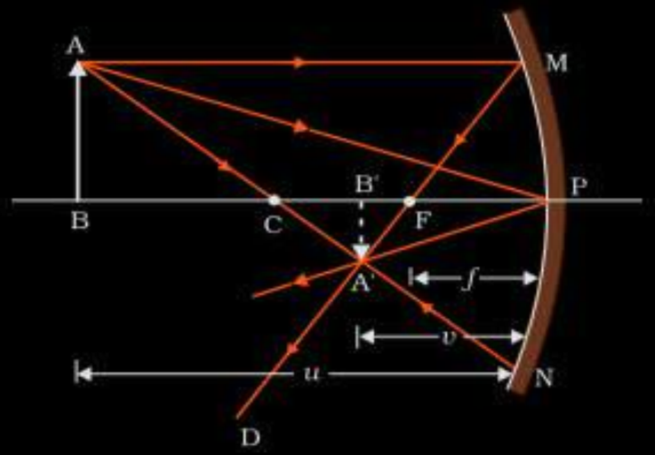
For object P $$m = \frac{h_2}{h_1} = \frac{f}{f - u_1}$$ For Object Q	½
$$m' = \frac{h_2'}{h_1'} = \frac{f}{f - u_2}$$	½
Now $h_1 = 3h_1'$; $h_2 = h_2'$; $f = -20\ cm$; $u_1 = -50\ cm$	
$$\frac{m}{m'} = \frac{h_2}{h_1} \times \frac{h_1'}{h_2'} = \frac{f - u_2}{f - u_1}$$	½
$$\frac{m}{m'} = \frac{1}{3} = \frac{-20 - u_2}{-20 + 50}$$	1
$10 = -20 - u_2 \Rightarrow\ u_2 = -30\ cm$	½

30. (a) Using the necessary ray diagram, derive the mirror formula for a concave mirror.

(b) In the magnified image of a measuring scale (with equidistant markings) lying along the principal axis of a concave mirror, the markings are not equidistant. Explain. *3*

CBSE 2020

a) Ray diagrams for concave mirror



½

Derivation of Mirror Formula

From the diagram,

$\Delta A'B'F$ & ΔMPF are similar

$$\therefore \frac{B'A'}{PM} = \frac{B'F}{FP}$$

$$\frac{B'A'}{BA} = \frac{B'F}{FP} \ (\because PM = AB) - - - - - - - - eq1$$

½

Since

$$\angle APB = \angle A'PB'$$

$\Delta A'B'P$ & ΔABP are also similar

$$\frac{B'A'}{BA} = \frac{B'P}{BP} \quad - - - - - - - - - - eq\ 2$$

½

Comparing eq. 1 and eq. 2

$$\frac{B'P}{BP} = \frac{B'P - FP}{FP}$$

As per the sign convention

$B'P = -v, \quad FP = -f,$ BP = -u

$$\frac{-v + f}{-f} = \frac{-v}{-u} = \frac{v}{u}$$

$$-vu + uf = -vf$$

Dividing by uvf

$$\Rightarrow \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

1

½

b) Magnification is different for different object distances

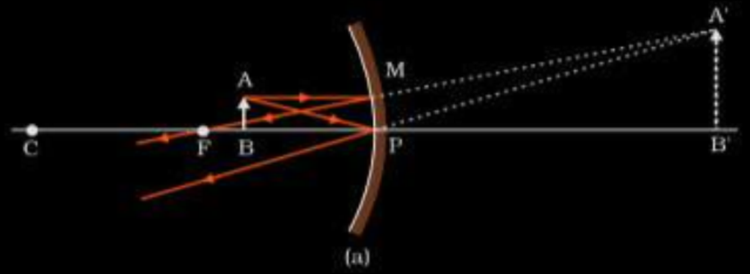
25. An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed. CBSE 2020 *2*

25. An object is kept 20 cm in front of a concave mirror of radius of curvature 60 cm. Find the nature and position of the image formed. CBSE 2020 *2*

The focal length $f = \frac{-R}{2} = -30\ cm\ u = -20\ cm$	½
$\frac{1}{V} + \frac{1}{u} = \frac{1}{f}$	½
$\therefore \frac{1}{V} - \frac{1}{20} = -\frac{1}{30} \Rightarrow \frac{1}{V} = -\frac{1}{30} + \frac{1}{20}$	½
$\therefore V = +60\ cm$	
Nature of image: virtual, erect and magnified	½

(a) An object is placed in front of a concave mirror. It is observed that a virtual image is formed. Draw the ray diagram to show the image formation and hence derive the mirror equation $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$.

CBSE 2020



1

$\Delta ABP \sim A'B'P$

$$\frac{A'B'}{AB} = \frac{PB'}{PB} - - - - - - - - - - 1$$

½

Also $\Delta A'B'F \sim MNP$ (for small curvature)

$$\therefore \frac{A'B'}{MP} = \frac{B'F}{PF}$$

$$\frac{A'B'}{AB} = \frac{B'F}{PF} - - - - - - - - - - - - 2$$

From 1 and 2

$$\frac{PB'}{PB} = \frac{B'F}{PF} - - - - - - - - - - - - 3$$

½

$$\frac{PB'}{PB} = \frac{B'P + PF}{PF} - - - - - - - - -4$$

½

$PB = -u \qquad PB' = v \qquad PF = -f$

$$\frac{v}{-u} = \frac{v - f}{-f}$$

$$-vf = -vu + uf$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

½

An object is kept in front of a concave mirror of focal length 15 cm. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.

CBSE 2019

An object is kept in front of a concave mirror of focal length 15 cm. The image formed is real and three times the size of the object. Calculate the distance of the object from the mirror.

CBSE 2019

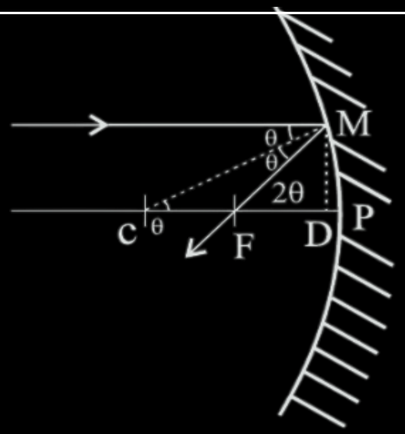
Here, $m = +3$ and $f = -15$ cm	
$m = \frac{v}{u} = 3 \quad \therefore \quad v = 3u$	½
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	½
$\frac{1}{-15} = \frac{1}{3u} + \frac{1}{u}$	½
$\Rightarrow \quad u = -20$ cm	½

For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

CBSE 2019

For paraxial rays, show that the focal length of a spherical mirror is one-half of its radius of curvature.

CBSE 2019



1

From ray diagram

$$\tan\theta = \frac{MD}{CD} \quad \text{and} \quad \tan 2\theta = \frac{MD}{FD}$$

½

for small θ, $\tan\theta \cong \theta$, $\tan 2\theta \cong 2\theta$

$$\therefore \frac{MD}{FD} = 2\frac{MD}{CD}$$

$$FD = \frac{CD}{2} \cong \frac{CP}{2}$$

½

$$\therefore \quad f = \frac{R}{2}$$

(a) With the help of a ray diagram, show how a concave mirror is used to obtain an erect and magnified image of an object.

(b) Using the above ray diagram, obtain the mirror formula and the expression for linear magnification.

CBSE 2018

(a) Calculate the distance of an object of height h from a concave mirror of radius of curvature 20 cm, so as to obtain a real image of magnification 2. Find the location of image also.

(b) Using mirror formula, explain why does a convex mirror always produce a virtual image.

CBSE 2016

Example 9.1 Suppose that the lower half of the concave mirror's reflecting surface in Fig. 9.5 is covered with an opaque (non-reflective) material. What effect will this have on the image of an object placed in front of the mirror?

Example 9.1 Suppose that the lower half of the concave mirror's reflecting surface in Fig. 9.5 is covered with an opaque (non-reflective) material. What effect will this have on the image of an object placed in front of the mirror?

Solution You may think that the image will now show only half of the object, but taking the laws of reflection to be true for all points of the remaining part of the mirror, the image will be that of the whole object. However, as the area of the reflecting surface has been reduced, the intensity of the image will be low (in this case, half).

Example 9.2 A mobile phone lies along the principal axis of a concave mirror, as shown in Fig. 9.7. Show by suitable diagram, the formation of its image. Explain why the magnification is not uniform. Will the distortion of image depend on the location of the phone with respect to the mirror?

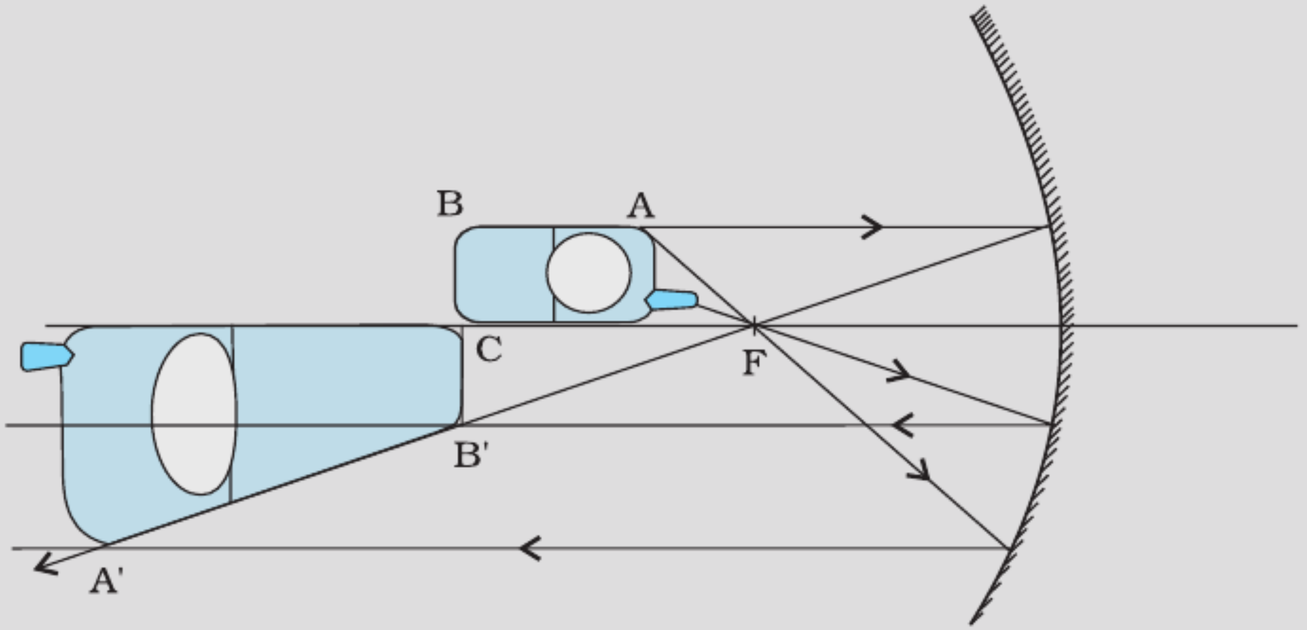


FIGURE 9.7

Solution

The ray diagram for the formation of the image of the phone is shown in Fig. 9.7. The image of the part which is on the plane perpendicular to principal axis will be on the same plane. It will be of the same size, i.e., B′C = BC. You can yourself realise why the image is distorted.

Example 9.3 An object is placed at (i) 10 cm, (ii) 5 cm in front of a concave mirror of radius of curvature 15 cm. Find the position, nature, and magnification of the image in each case.

Example 9.3 An object is placed at (i) 10 cm, (ii) 5 cm in front of a concave mirror of radius of curvature 15 cm. Find the position, nature, and magnification of the image in each case.

Solution

The focal length $f = -15/2$ cm $= -7.5$ cm

(i) The object distance $u = -10$ cm. Then Eq. (9.7) gives

$$\frac{1}{v}+\frac{1}{-10}=\frac{1}{-7.5}$$

or $$v=\frac{10\times7.5}{-2.5}=-30 \text{ cm}$$

The image is 30 cm from the mirror on the same side as the object.

Also, magnification $m = -\frac{v}{u}=-\frac{(-30)}{(-10)}=-3$

The image is magnified, real and inverted.

(ii) The object distance $u = -5$ cm. Then from Eq. (9.7),

$$\frac{1}{v}+\frac{1}{-5}=\frac{1}{-7.5}$$

or $$v=\frac{5\times7.5}{(7.5-5)}=15 \text{ cm}$$

This image is formed at 15 cm behind the mirror. It is a virtual image.

Magnification $m = -\frac{v}{u}=-\frac{15}{(-5)}=3$

The image is magnified, virtual and erect.

Example 9.4 Suppose while sitting in a parked car, you notice a jogger approaching towards you in the side view mirror of $R = 2$ m. If the jogger is running at a speed of 5 m s^{-1}, how fast the image of the jogger appear to move when the jogger is (a) 39 m, (b) 29 m, (c) 19 m, and (d) 9 m away.

Solution

From the mirror equation, Eq. (9.7), we get

$$v = \frac{fu}{u - f}$$

For convex mirror, since R = 2 m, f = 1 m. Then

for u = –39 m, $v = \frac{(-39)\times 1}{-39-1} = \frac{39}{40}$ m

Since the jogger moves at a constant speed of 5 m s^{-1}, after 1 s the position of the image v (for u = –39 + 5 = –34) is (34/35)m.

The shift in the position of image in 1 s is

$$\frac{39}{40} - \frac{34}{35} = \frac{1365-1360}{1400} = \frac{5}{1400} = \frac{1}{280}\text{ m}$$

Therefore, the average speed of the image when the jogger is between 39 m and 34 m from the mirror, is (1/280) m s^{-1}

Similarly, it can be seen that for u = –29 m, –19 m and –9 m, the speed with which the image appears to move is

$\frac{1}{150}$ m s^{-1}, $\frac{1}{60}$ m s^{-1} and $\frac{1}{10}$ m s^{-1}, respectively.

Although the jogger has been moving with a constant speed, the speed of his/her image appears to increase substantially as he/she moves closer to the mirror. This phenomenon can be noticed by any person sitting in a stationary car or a bus. In case of moving vehicles, a similar phenomenon could be observed if the vehicle in the rear is moving closer with a constant speed.

9.1 A small candle, 2.5 cm in size is placed at 27 cm in front of a concave mirror of radius of curvature 36 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? Describe the nature and size of the image. If the candle is moved closer to the mirror, how would the screen have to be moved?

9.2 A 4.5 cm needle is placed 12 cm away from a convex mirror of focal length 15 cm. Give the location of the image and the magnification. Describe what happens as the needle is moved farther from the mirror.

9.15 Use the mirror equation to deduce that:

(a) an object placed between f and $2f$ of a concave mirror produces a real image beyond $2f$.

(b) a convex mirror always produces a virtual image independent of the location of the object.

(c) the virtual image produced by a convex mirror is always diminished in size and is located between the focus and the pole.

(d) an object placed between the pole and focus of a concave mirror produces a virtual and enlarged image.

[*Note:* This exercise helps you deduce algebraically properties of images that one obtains from explicit ray diagrams.]